An RL-Based Adaptive Traffic Signal Control System

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Abstract

Traffic signal control is important for reducing traffic congestion, but fixed-timing strategies cannot adjust to changing traffic conditions. This project uses reinforcement learning to train a PPO agent for a single-intersection SUMO environment. The agent is tested under different reward functions, environment settings, and hyperparameters. Experiments with queue-based, pressure-based, and hybrid rewards show that while simple rewards can speed up early learning, they often lead to unstable behavior. The hybrid reward performs the most consistently across all tests. Overall, the PPO agent outperforms the fixed-duration baseline.

1 Introduction

As cities grow, traffic demand often exceeds road capacity, leading to congestion and delays. Fixed-time traffic signals cannot adjust to changing traffic patterns, making them less effective in busy urban settings. Because signal control is a sequential decision problem driven by real-time conditions, reinforcement learning (RL) offers a way to adapt signal phases dynamically. Prior work has shown that RL can improve traffic signal control [1, 2], and an RL agent can adjust decisions based on observed traffic states to handle variability more effectively.

2 Preliminaries and Problem Formulation

Optimizing traffic signal control plays a crucial role in reducing congestion and improving mobility. In this project, we frame traffic signal control as a sequential decision-making problem and apply RL to learn effective control strategies. The goal is to train an RL agent that can improve fixed-time or other heuristic controllers by dynamically selecting signal phases in response to evolving traffic conditions. Through repeated interaction with the environment, the agent aims to minimize the amounts related to congestion.

We model the traffic signal control problem as a Markov Decision Process (MDP). At each discrete time step, the environment is characterized by a state capturing relevant traffic conditions, the agent selects an action indicating the desired signal phase, and the traffic dynamics evolve accordingly. A reward function provides immediate feedback on the effectiveness of the control and the agent seeks to learn a policy that maximizes the expected discounted return.

In traffic systems, various congestion indicators are used to quantify imbalance or inefficiency. Two important concepts are movement pressure and intersection pressure, which originate from transportation theory and have been widely adopted in RL-based traffic control formulations:

- Pressure of movement: The difference between the number of vehicles entering $x(e)$ and leaving $x(l)$ the intersection on each lane L_i.

$$w(L_i) = x(e) - x(l) \tag{1}$$

- Pressure of intersection: The total absolute pressure across all its movements. It can illustrate the imbalance of an intersection — the larger the value, the more congested the intersection.

$$P_t = \left| \sum_i w(L_i) \right| \tag{2}$$

3 Design

3.1 Environment Setup

To simulate a single four-way traffic intersection, we used the Simulation of Urban Mobility (SUMO) microscopic traffic simulator [3]. At each time step the agent selects a signal phase, the simulator advances, and updated traffic states are returned. To evaluate robustness, we injected zero-mean Gaussian noise into all of the observed states, simulating realistic sensor measurement uncertainty.

3.2 State Space

Each state represents the current situation of this intersection and all lanes. We used several indicators extracted from the `sumo-rl` environment to illustrate the real-time environment situation. Vehicles refer to vehicles visible in the intersection window.

- Inbound and Outbound Vehicle Counts
- Queue Length
- Average waiting time for vehicles
- Average speed of vehicles
- Pressure of the intersection
- Traffic-light phase, as shown in Table 1

3.3 Action Space

The action space consists of a discrete set of eight predefined traffic-signal phases specified in the SUMO traffic-light logic. At each time step, the agent selects an action

$$a_t \in \{0, 1, 2, 3, 4, 5, 6, 7\},$$

corresponding exactly to one of the eight predefined traffic-light phases as shown in Table 1. Notes that Northbound and Westbound are separated to accommodate asymmetric traffic patterns and independent movement control. SUMO automatically handles yellow and safety transitions, so the agent's role is to choose the next phase rather than control timing. This keeps the action space compact while enabling adaptive signal control.

This formulation allows the agent to learn phase-level control without imposing any directional or structural constraints, enabling more expressive traffic-signal policies compared to fixed two-phase designs.

3.4 Reward Function

We evaluate three reward functions commonly used in traffic signal control:

Table 1: Predefined traffic-light phases in the SUMO signal program

Phase Index	Description
0	East–West through green
1	East–West through yellow
2	Northbound through green
3	Northbound yellow
4	South–North through green
5	South–North through yellow
6	Westbound through green
7	Westbound yellow

- Queue-based reward:
  $$r_t = -Q_t \tag{3}$$
  This function penalizes the instantaneous total queue length Q_t. It directly encourages the agent to reduce queued vehicles and is widely used due to its simplicity and interpretability.
- Pressure-based reward:
  $$r_t = -P_t \tag{4}$$
  This function penalizes the intersection pressure P_t, defined before as the sum of movement imbalances. It encourage the .
- Combined reward (queue, pressure, speed, switching penalty):
  $$r_t = c_1(Q_{t-1} - Q_t) - c_2 Q_t - c_3 P_t - c_4 S_t + c_5 T_t \tag{5}$$
  This composite reward integrates multiple congestion indicators. It rewards queue reduction $Q_{t-1} - Q_t$, penalizes queue size Q_t and pressure P_t, discourages excessive phase switching S_t and rewards throughput T_t. The weighted structure allows more nuanced control behavior and facilitates evaluating how different components influence learning. In our experiments, we use the coefficients $(c_1, c_2, c_3, c_4, c_5) = (1.0,\ 0.3,\ 0.15,\ 0.05,\ 0.005)$, chosen empirically based on preliminary runs.

4 Methodology

4.1 Baseline

We implement a fixed-time traffic-signal controller as the baseline. This controller does not incorporate learning or real-time traffic information. Instead, it follows a predetermined sequence of signal phases, where each phase is activated for a constant and predefined duration.

Internally, the controller maintains two variables: the current phase index and the number of steps the phase has been active. At each time step, if the phase has not yet reached its preset duration, the signal remains unchanged. Once the duration is exceeded, the controller advances to the next phase in the sequence and resets the counter.

This baseline serves as a reference for evaluating the adaptability and effectiveness of the PPO-based controller under dynamic traffic conditions.

4.2 PPO

The control strategy is learned using the Proximal Policy Optimization framework, which adopts an Actor–Critic structure. The policy network maps traffic states to probability distributions over signal phases, while a parallel value estimator predicts long-term returns. The learning process alternates between interaction with the simulation environment and policy updates.

During interaction, the agent collects batches of sequential experience that include observed states, selected actions, immediate traffic performance feedback, and termination signals. These samples are used to compute advantage estimates through generalized advantage estimation, enabling more stable and sample-efficient learning. The policy is updated by optimizing a clipped objective that constrains overly large deviations from the previous policy, while the value estimator is trained to improve state-value predictions. An entropy term encourages exploration, and gradient clipping enhances training stability. Throughout training, traffic indicators such as delay, queuing levels, pressure, speed, and throughput are monitored to assess learning progress.

4.3 Pipeline

In this project, we followed a structured pipeline to develop and evaluate an intelligent traffic-signal control system. We began by constructing the single-intersection SUMO environment encapsulating with the necessary modification module and built a heuristic fixed-duration baseline and a PPO-based RL agent capable of dynamic and state-dependent signal control. After conducting hyperparameter tuning to ensure stable convergence of PPO, we carried out a series of core experiments organized to examine the effects of different reward functions and evaluated the robustness of the agent under sensor noise with both the RL agent and heuristic baselines. Finally, we generated quantitative tables in terms of averages and rolling metrics and visualizations to analyze and compare performance across all configurations.

5 Numerical Experiments

The experimental setup and results of the traffic-signal control system will be specified in this section. All experiments were conducted on the single-intersection SUMO network provided in our environment files defined in the corresponding configurations. The PPO-based RL agents are evaluated under different hyper-parameters heuristic baselines and environment modifications. The selected potential best-performing agents and baselines eventually will be tested with under the original unmodified environment.

5.1 Hyperparameter Tuning

The hyper-parameter tuning focuses on the parameters that are highly affect the converge speed and the smoothness in the long-term stability of PPO learning process, which inclues learning rate, batch size, GAE γ and the policy clip ratio. By reviewing the multiple configurations, the results indicates that all the parameters setting are having difficulty on making a smooth learning curves and avoiding the bouncing up and down. In the end, the parameters produce the best average on rewards value will be selected as the finalized configuration for the later experiments, which are: learning rate 3×10^{-4} , mini-batch size 64, GAE $\gamma = 0.99$ and the policy clip ratio.

5.2 Comparison Across Modifications and Controllers

To assess the stability of the system and understand how various design choices influence control performance, we evaluate the PPO agent under eight configurations. The detailed parameters of configuration for each experiment can be accessed in the code files. To further evaluate the best model, we chose to use 100-episode travel time (lower is better) as the primary performance metric, following prior studies such as PressLight (Wei et al., 2019).

Table 2: Metrics Comparsion

experiment_id	avg_reward	avg_waiting_time	avg_queue_length	avg_pressure
1	-9.200	12.400	15.326	29.318
2	-13.743	13.651	13.743	51.065
3	-103.240	42.199	52.929	103.240
4	-10.552	11.675	18.202	32.521
5	-9.885	13.841	9.885	55.261
6	-64.744	6.608	22.709	64.744
7	-46.168	51.505	79.679	147.896
8	-9.666	11.654	15.637	31.879

From the average waiting time comparison curves of all experiments in Figure 1, it can be observed that reward function design, perception noise, and controller type have a significant impact on traffic signal control performance. Among all reinforcement learning-based models, the default reward showed relatively low and stable waiting times in a noise-free environment (exp1) as well as in the second-stage experiment under the same configuration (exp8). Rewards focused on queue length

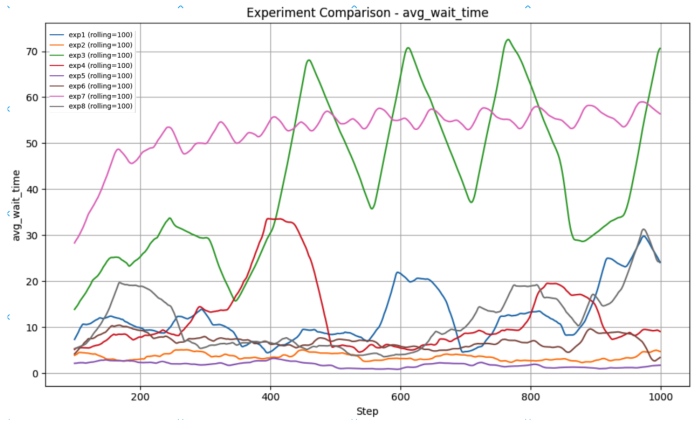

Figure 1: Final evaluation of the trained PPO agent in the original SUMO environment.

demonstrated strong stability in both noise-free (exp2) and noisy environments (exp5), with exp5 showing the lowest average waiting time across all experiments. This suggests that queue length as a state feature is less affected by noise, allowing such rewards to drive robust behavior even under inaccurate perception conditions. In contrast, reward functions primarily targeting traffic pressure (exp3 and exp6) exhibited significantly higher waiting times with pronounced fluctuations. The root cause is that the pressure metric is highly sensitive to instantaneous inflow and outflow differences, causing the policy to over-rely on immediate signals and frequently switch phases, introducing unstable control behavior. Even in a noise-free scenario (exp3), large fluctuations are observed. Similar observation can be seen in the exp6). The heuristic baseline (exp7) performed the least favorably, with overall waiting times consistently high and exhibiting periodic oscillations, reflecting the lack of adaptability of fixed-phase logic in the face of dynamic traffic flows, resulting in continuously accumulating queues.

6 Discussion

Experimental results show that, under different environmental settings, PPO-based adaptive control consistently outperforms fixed-phase heuristic baseline strategies by significantly reducing average waiting times and improving traffic flow stability. Our study also found that the design of the reward function plays an important role in enhancing policy learning. From the trend, we can see that the reward formulation combined features effectively enhance training stability and long-term performance and the pressure-based reward method is sensitive to instantaneous flow fluctuations and implies unstable control with frequent bouncing up and down. In contrast, the queue-length-based reward experiment remains robust in noisy environments which indicates that it has greater applicability in real-world scenarios with perception errors. In summary, although PPO demonstrates clear performance advantages, its training process is still sensitive to hyperparameter tuning and this study was only evaluated in a single intersection scenario, limiting the generalization of the conclusions.

7 Conclusion

This project revealed how sensitive PPO is to design choices such as reward functions, noise adding, and hyperparameter settings. We observed that small changes in these components can noticeably affect training stability, convergence speed, and the final behavior of the agent. Among the tested options, the hybrid reward led to the most stable and balanced control performance, consistently outperforming the fixed-time baseline. Future work may explore more advanced reward formulations beyond simple linear combinations and experiment with alternative traffic-light phase designs to support more flexible control.

Assentation of Teamwork

This project was completed through collaborative effort among all team members, with each contributor responsible for different components of the system. The SUMO environment and baseline controller were implemented by Yong, who also conducted the experiments and evaluation. The reward design was developed by Zimo, while the PPO algorithm implementation was carried out by Hengyu. The overall system pipeline, including integration of the environment, agent, and evaluation modules, was jointly assembled by Yong and Zimo, who also explored alternative action designs. Most sections of the written report were prepared by Yong and Zimo, with Hengyu contributing the PPO and baseline descriptions. The final compilation, editing, and refinement of the report were completed by Zimo. Based on the division of work, the contributions are estimated as follows: Yong Chen (40%), Zimo Zhang (40%), and Hengyu Liu (20%)

References

[1] Wei, H., Chen, C., Zheng, G., Wu, K., Gayah, V., Xu, K., & Li, Z. (2019). *PressLight. Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 1290–1298. https://doi.org/10.1145/3292500.3330949

[2] Wei, H., Xu, N., Zhang, H., Zheng, G., Zang, X., Chen, C., Zhang, W., Zhu, Y., Xu, K., & Li, Z. (2019). *Colight. Proceedings of the 28th ACM International Conference on Information and Knowledge Management*, 1913–1922. https://doi.org/10.1145/3357384.3357902

[3]López, P., Behrisch, M., Bieker-Walz, L., Erdmann, J., Wang, Y., & Weber, M. (2018). *Microscopic Traffic Simulation using SUMO. In 2018 21st IEEE International Conference on Intelligent Transportation Systems (ITSC).*,